

O1 Baseline Comparison

Fixed 474-GM test set | Full-data protocol | Batch size 64 | 50 epochs

Executive Summary

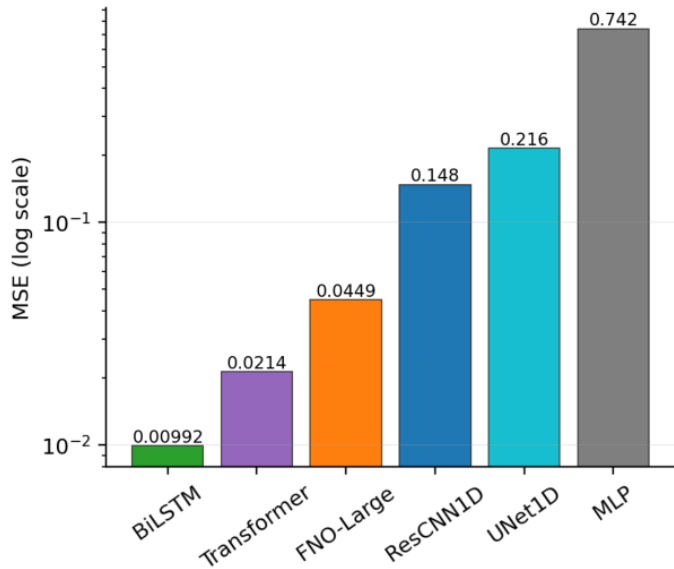
- BiLSTM achieved the lowest test MSE (0.0099) and highest R2 (0.9872), but required the longest training time (7.51 h) and largest memory footprint (15.61 GB).
- The stabilized pre-norm Transformer ranked second in MSE (0.0214) with substantially lower memory than BiLSTM (2.54 GB).
- FNO-Large was resource efficient but underperformed BiLSTM/Transformer on this O1 fixed 474-GM test split (MSE 0.0449).
- The original Transformer run collapsed to a mean-prediction regime and was excluded after replacing it with a standard pre-norm Transformer and a lower learning rate.

Leaderboard (fixed 474-GM test MSE)

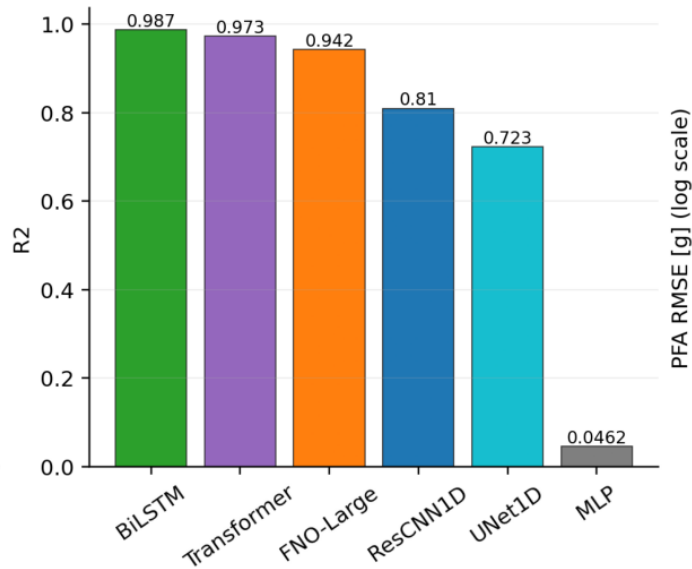
Model	MSE	R2	PFA RMSE [g]	Train h	GPU GB
BiLSTM	0.0099	0.9872	0.0182	7.51	15.61
Transformer	0.0214	0.9725	0.0231	2.34	2.54
FNO-Large	0.0449	0.9423	0.0330	2.16	3.30
ResCNN1D	0.1481	0.8096	0.0547	6.29	3.49
UNet1D	0.2157	0.7227	0.1063	2.27	3.03
MLP	0.7420	0.0462	0.2464	1.59	0.24

Fixed 474-GM Test Set Accuracy

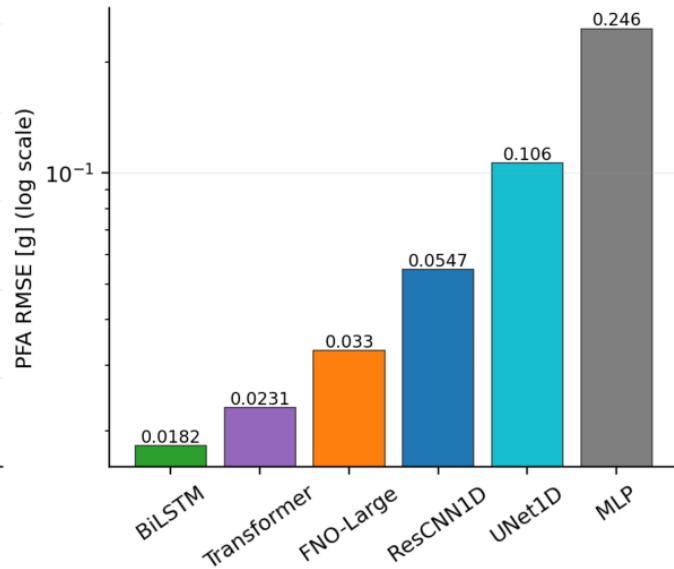
Test MSE



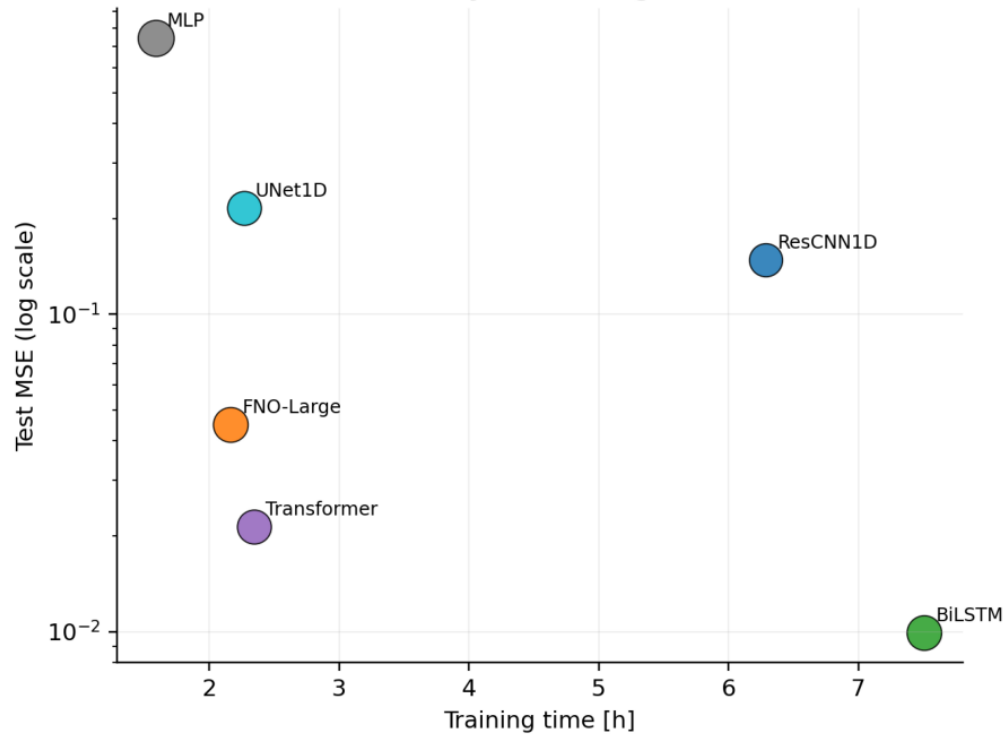
Test R2



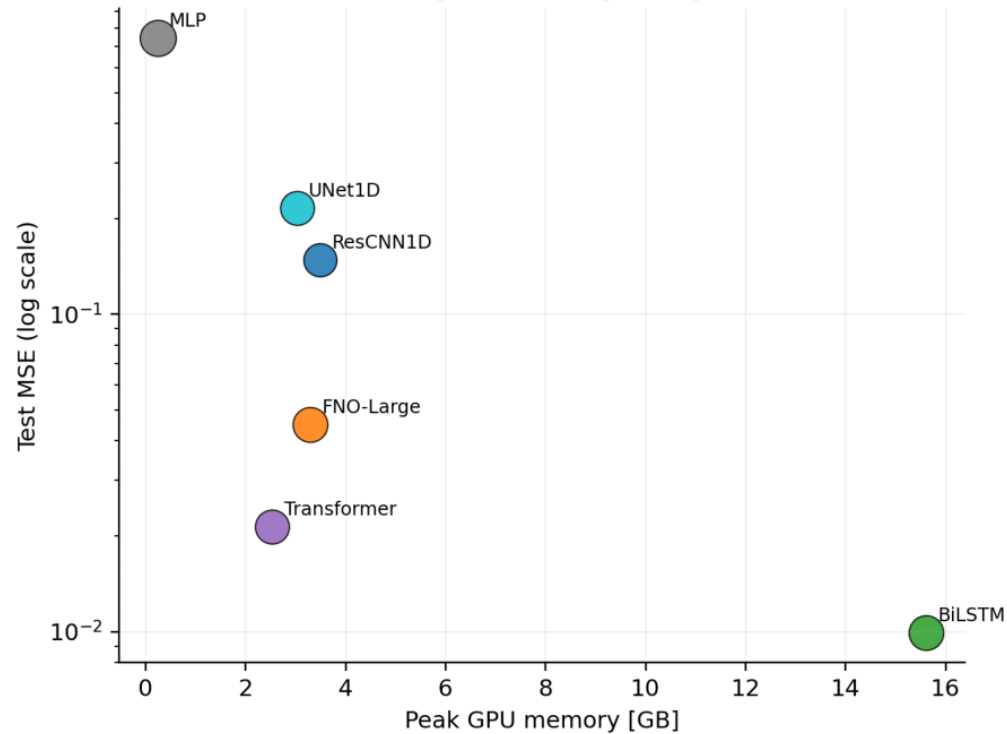
PFA RMSE



Accuracy vs Training Cost

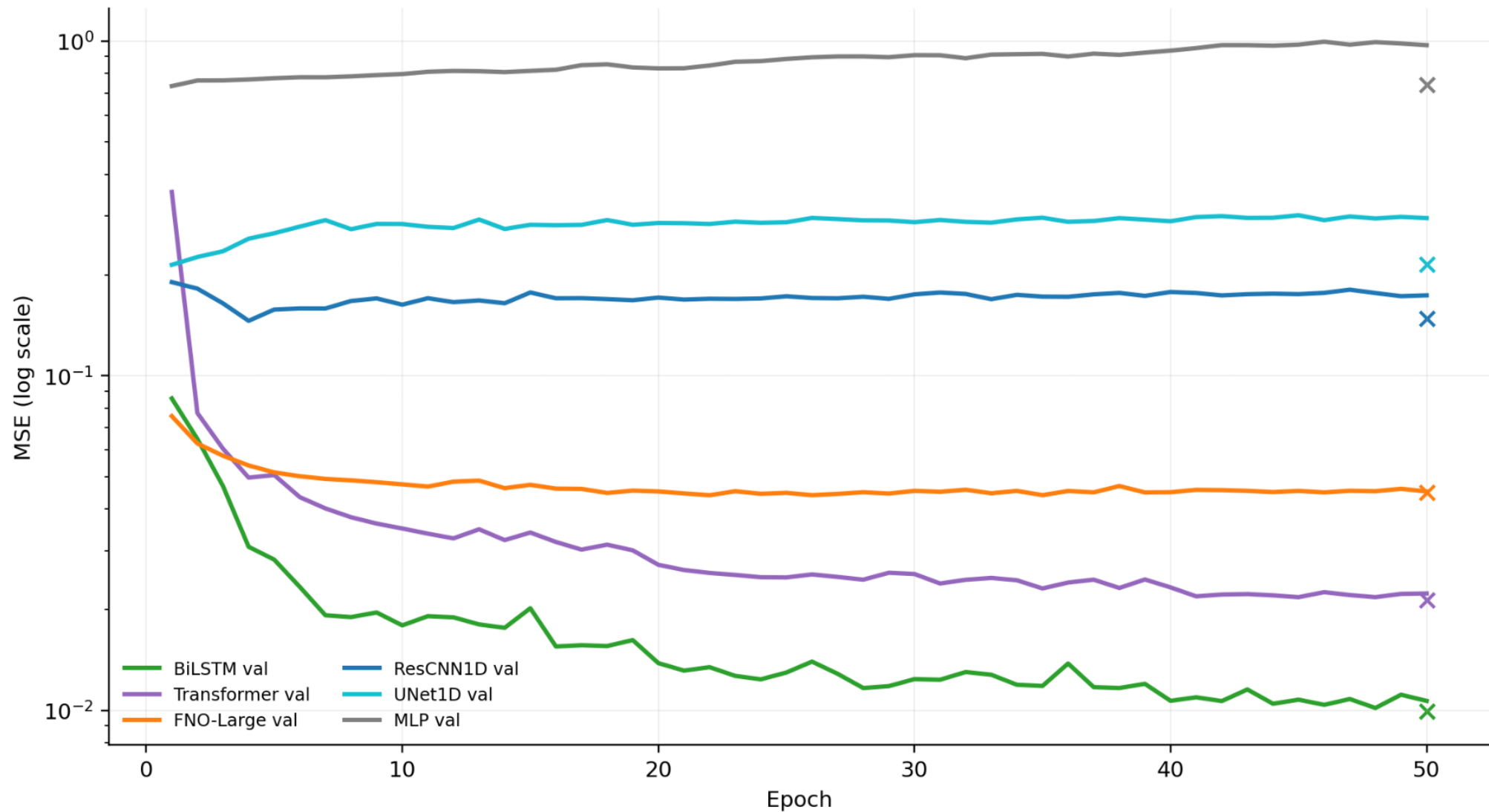


Accuracy vs Memory Footprint



Bubble area scales with number of trainable parameters.

Validation MSE Curves with Final Test MSE Markers



Normalized Accuracy and Resource Profile

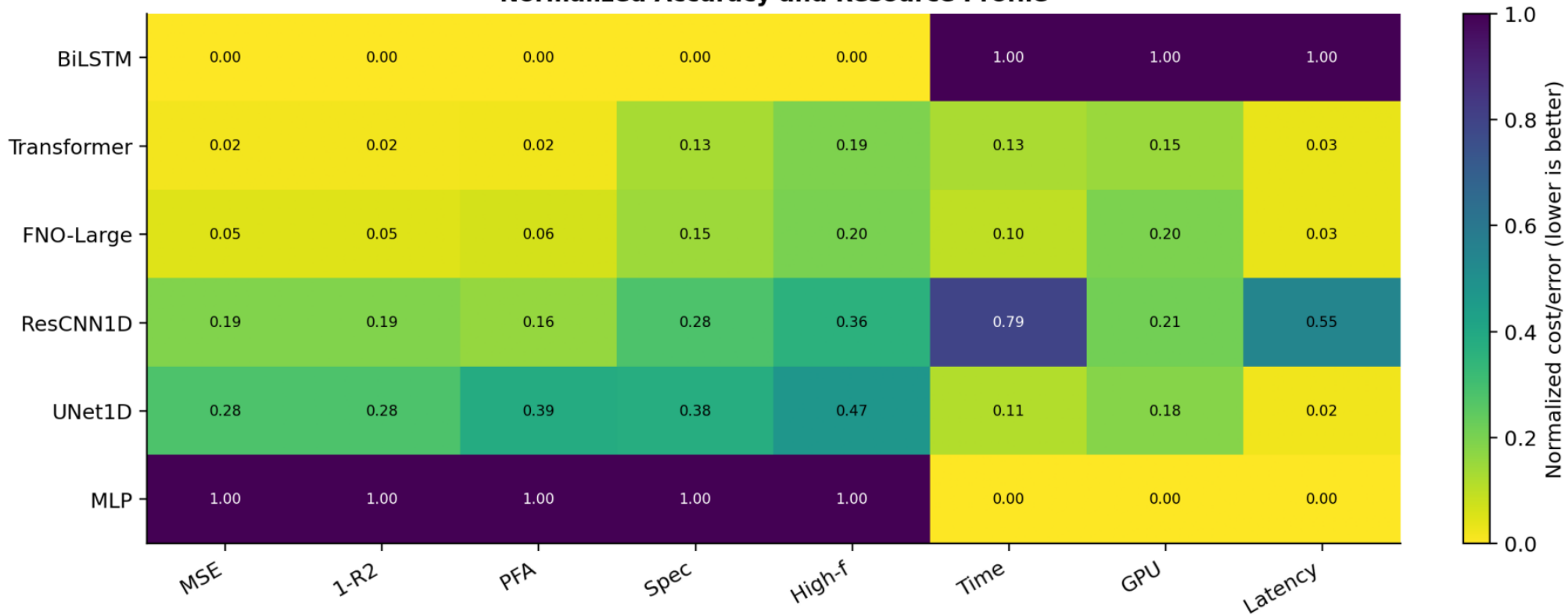


Table 1. Fixed 474-GM Test Results and Resource Costs

Model	MSE ↓	R2 ↑	PFA RMSE ↓	Spec ↓	High-f ↓	Train h ↓	GPU GB ↓	Latency ms ↓
BiLSTM	0.0099	0.9872	0.0182	0.185	0.222	7.51	15.61	1.049
Transformer	0.0214	0.9725	0.0231	0.325	0.391	2.34	2.54	0.106
FNO-Large	0.0449	0.9423	0.0330	0.342	0.398	2.16	3.30	0.108
ResCNN1D	0.1481	0.8096	0.0547	0.484	0.534	6.29	3.49	0.607
UNet1D	0.2157	0.7227	0.1063	0.588	0.633	2.27	3.03	0.096
MLP	0.7420	0.0462	0.2464	1.255	1.091	1.59	0.24	0.075